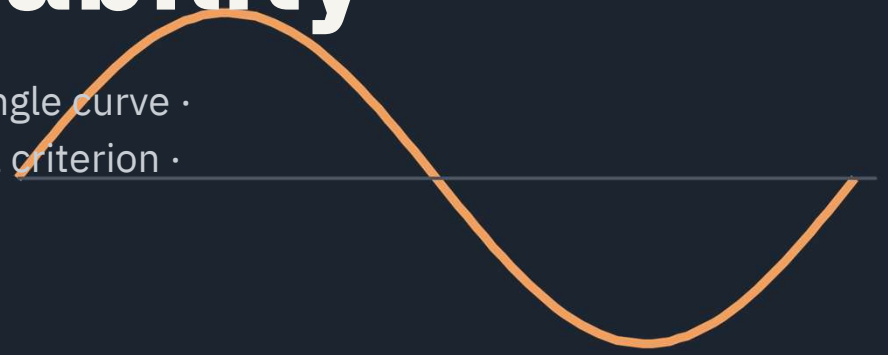
POWER SYSTEMS ENGINEERING

Power System Stability

Steady-state, dynamic and transient stability · power angle curve · steady-state stability limit · swing equation · equal area criterion · improving stability



INTRODUCTION

What is power system stability?

Power system stability is the ability of a power system to return to normal, stable operation after it has been disturbed.

> **An analogy**
>
> Think of the generators as runners tied together with elastic ropes. After a push they wobble, but if the ropes hold they keep running together. If a rope snaps, a runner falls out of step.

INTRODUCTION

Stable, unstable and what disturbs the system

In simple words

All synchronous generators must keep rotating together at the same electrical speed. If they stay in step after a disturbance, the system is stable.

Instability

One or more machines fall out of step with the rest of the system (pole slipping). This is called loss of synchronism.

Typical disturbances

Short-circuit faults, sudden changes in load, loss of a generator or of a transmission line.

CLASSIFICATION

Three types of stability

Type	Disturbance	What we study	Key point
Steady-state	Slow, gradual increase in load	Maximum power without losing synchronism	Steady-state stability limit
Dynamic	Small, sudden changes	Do the oscillations die out quickly?	Often caused by control systems
Transient	Large and sudden: faults, loss of a line or generator	Does the rotor survive the large swing?	Decided within a few seconds

CLASSIFICATION

Assumptions in stability studies

Stability analysis mainly studies the rotor. To keep it simple, we assume:

Balanced system

The system and the disturbances are balanced, so a per-phase model is enough.

Small speed deviation

Machine speeds stay very close to synchronous speed.

Fast transients neglected

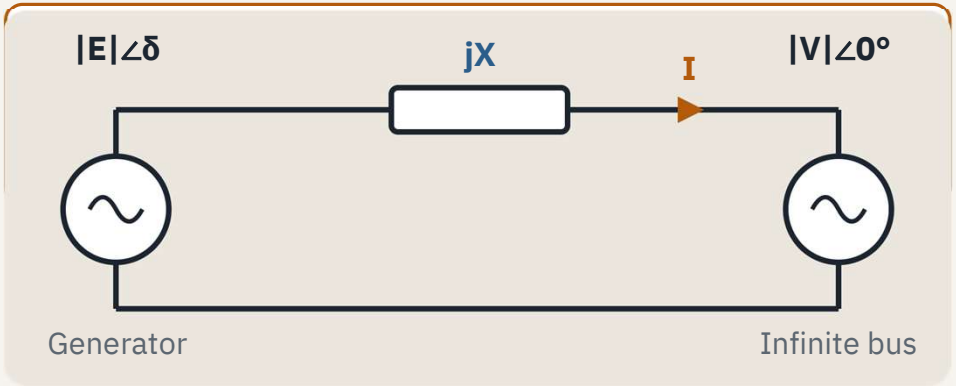
DC offset and high-frequency fault currents are ignored.

Network in steady state

Voltages, currents and powers come from power flow equations.

POWER ANGLE CURVE

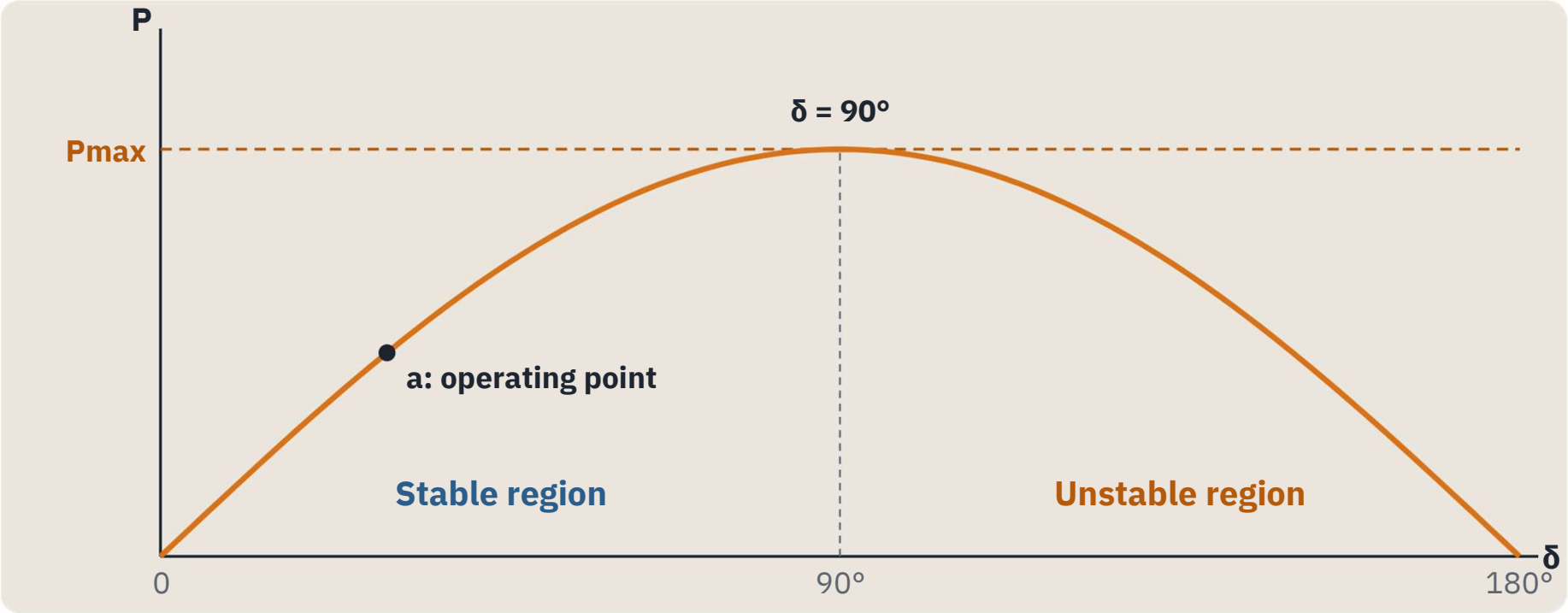
Power sent over a lossless line



1. Line current
   I = (|E|∠δ − |V|∠0°) / jX
2. Power sent is S = E I*; take its real part.
3. Real power depends on the angle δ between E and V
   P = (|E||V| / X) sin δ
4. Largest value at δ = 90°
   Pmax = |E||V| / X

POWER ANGLE CURVE

The power angle curve: P = Pmax sin δ



POWER ANGLE CURVE

Reading the power angle curve

Maximum power

Pmax = |E||V| / X, reached at δ = 90°.

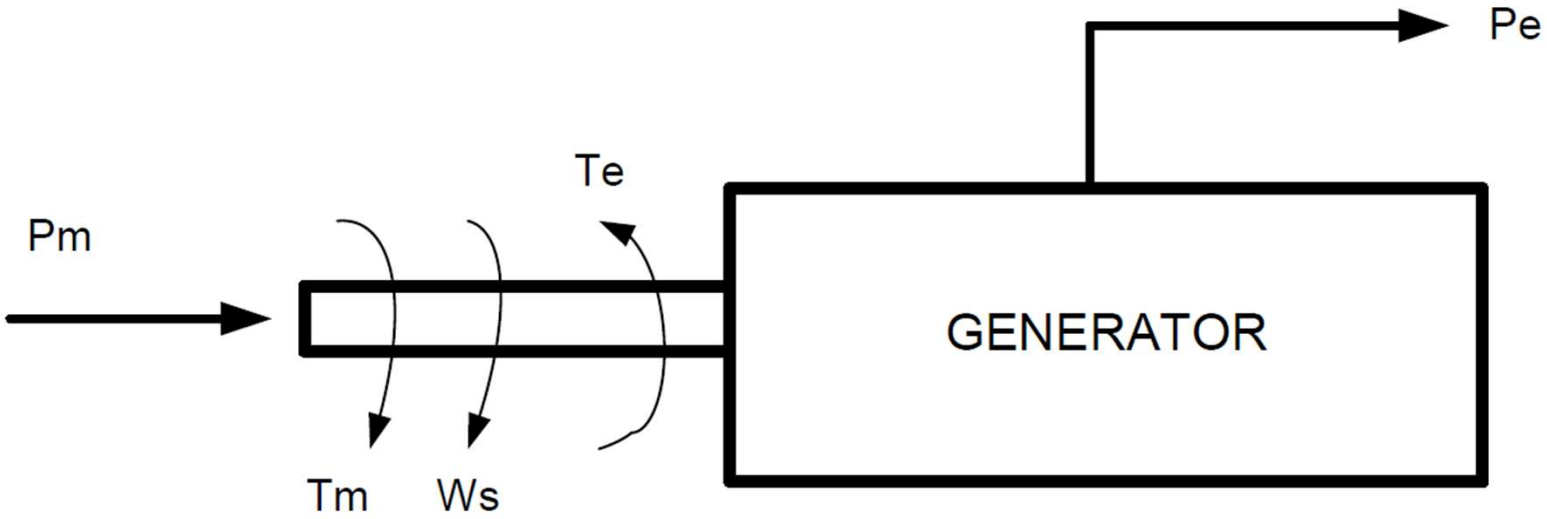
Stable: 0° to 90°

A larger δ gives more electrical power, so the machine can respond to extra input.

Unstable: 90° to 180°

A larger δ gives less power, so the rotor keeps speeding up and loses step.

Swing Equation:-



Consider a synchronous generator developing an electromagnetic torque T_e(and a corresponding electromagnetic power P_e) while operating at the synchronous speed w_s. If the input torque provided by the prime mover, at the generator shaft is T_i, then under steady state conditions (i.e., without any disturbance).

Derivation of Swing Equation

For a synchronous machine,

$$M = \frac{2GH}{\omega_{se}}$$

Since $\omega_{se} = 2\pi f$,

$$\boxed{M = \frac{GH}{\pi f}}$$

Symbols:

M – inertia coefficient

G – machine MVA rating

H – inertia constant

f – system frequency

Steady-State Condition

At steady state,

$$T_e = T_i$$

Multiplying by synchronous speed,

$$T_e\omega_s = T_i\omega_s$$

Hence,

$$\boxed{P_i - P_e = 0}$$

Symbols:

T_e – electromagnetic torque

T_i – input torque

P_i – mechanical input power

P_e – electrical output power

ω_s – synchronous speed

Accelerating Power

During steady state,

$$P_i = P_e$$

After a disturbance,

$$P_i \neq P_e$$

The difference is the **accelerating power**:

$$\boxed{P_a = P_i - P_e}$$

From rotational dynamics,

$$\text{Accelerating torque} = J\alpha$$

Converting torque to power and expressing rotor inertia by M,

$$\boxed{P_a = M\frac{d^2\theta_e}{dt^2}}$$

Including damping,

$$\boxed{P_i - P_e = M\frac{d^2\theta_e}{dt^2} + D\frac{d\theta_e}{dt}}$$

Symbols:

P_a – accelerating power

P_i – mechanical input power

P_e – electrical output power

J – rotor moment of inertia

α – angular acceleration

M – inertia coefficient

D – damping coefficient

θ_e – rotor electrical angle

t – time

Power Angle

Rotor angle relative to the synchronous reference:

$$\boxed{\delta = \theta_e - \omega_s t}$$

Since ω_s is constant,

$$\boxed{\frac{d^2\theta_e}{dt^2} = \frac{d^2\delta}{dt^2}}$$

Neglecting damping $(D = 0)$,

$$\boxed{M\frac{d^2\delta}{dt^2} = P_i - P_e}$$

Symbols:

δ – rotor power angle

θ_e – rotor electrical angle

ω_s – synchronous speed

Swing Equation

Substitute

$$M = \frac{GH}{\pi f}$$

Therefore,

$$\frac{GH}{\pi f}\frac{d^2\delta}{dt^2} = P_i - P_e$$

Dividing by machine rating G,

$$\boxed{\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_i - P_e \ (\text{pu})}$$

Symbols:

H – inertia constant

f – system frequency

δ – power angle

P_i – mechanical input power

P_e – electrical output power

POWER ANGLE CURVE · WORKED EXAMPLE

Example: using the power angle curve

Given: |E| = 1.2 pu, |V| = 1.0 pu, total reactance X = 0.6 pu. **Find** (i) Pmax, (ii) δ at P = 1.0 pu, and (iii)P at δ = 60°.

1. Maximum power
   Pmax = |E||V| / X = (1.2 × 1.0) / 0.6 = 2.0 pu
2. Power angle for P = 1.0 pu
   sin δ = 1.0 / 2.0 = 0.5 → δ = 30°
3. Power at δ = 60°
   P = 2.0 × sin 60° = 1.732 pu
4. Margin before the limit
   Pmax – P = 2.0 – 1.0 = 1.0 pu

ANSWER

Pmax = 2.0 pu

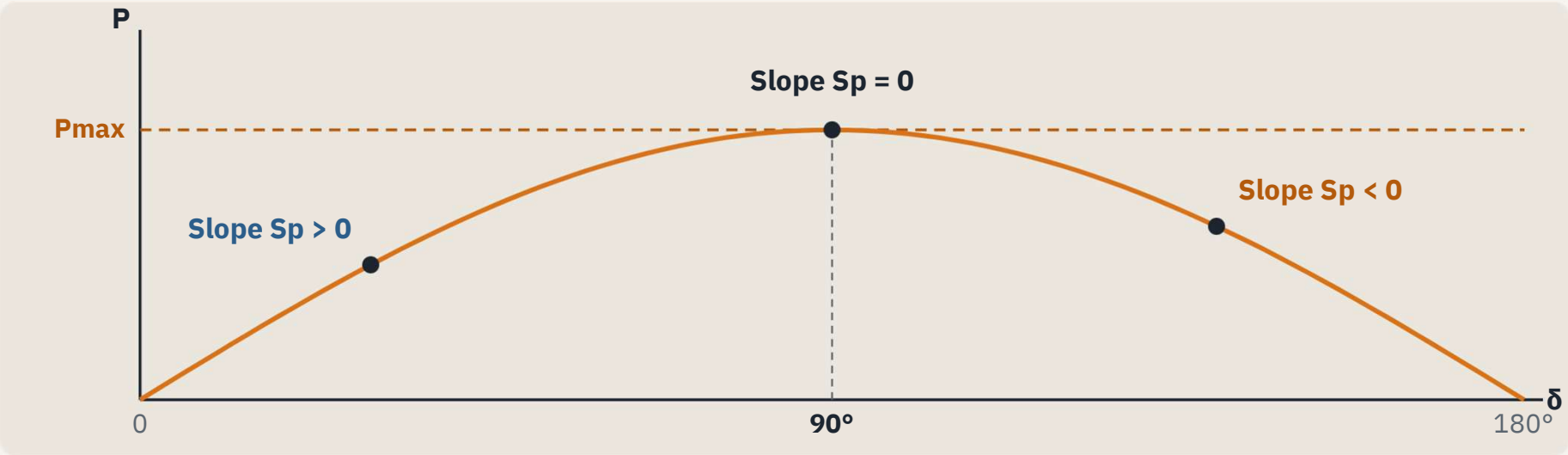
δ = 30°

P(60°) = 1.73 pu

STEADY-STATE STABILITY LIMIT

The steady-state stability limit

The **maximum power** that can be transferred without losing synchronism when the load is increased **slowly and gradually**.



STEADY-STATE STABILITY LIMIT

The synchronizing coefficient Sp

$$Sp = \frac{dP_e}{d\delta} = P_{max} \cos \delta$$

What it means

The extra power the machine sends out for a small increase in δ. Also called the stiffness.

Stable while Sp > 0

Sp is positive for δ < 90°, so the machine is steady-state stable there.

The limit

At δ = 90°, Sp = 0 and P = Pmax = |E||V| / X.

Rotor Angle at the Stability Limit

Taking the infinite-bus voltage as reference:

$$V = |V|\angle 0^\circ$$

Generator internal voltage:

$$E = |E|\angle \delta$$

At the steady-state stability limit:

$$\boxed{\delta = 90^\circ}$$

Therefore,

$$E = |E|(\cos 90^\circ + j \sin 90^\circ)$$

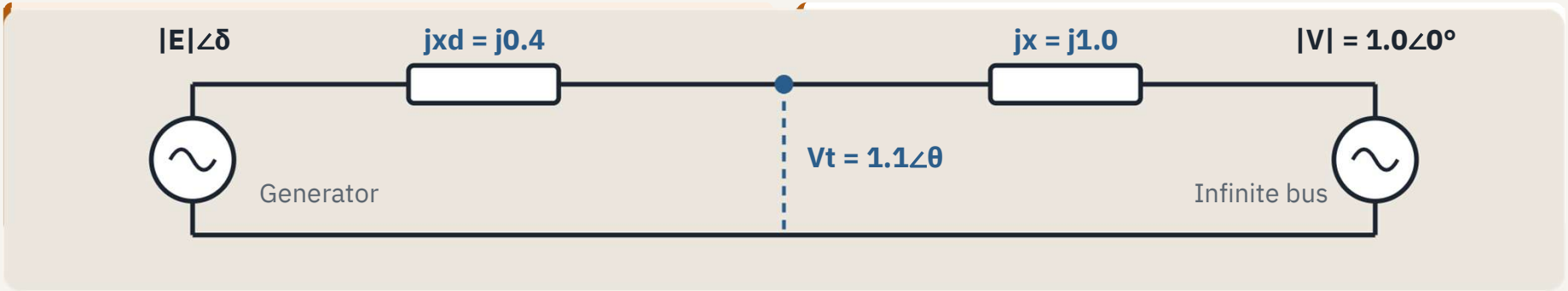
$$\boxed{E = j|E|}$$

Hence,

$$\boxed{\mathrm{Re}(E) = 0}$$

STEADY-STATE STABILITY LIMIT · WORKED EXAMPLE

Example: maximum steady-state power



STEADY-STATE STABILITY LIMIT · WORKED EXAMPLE

Solution (1): express E

1. Write E using the terminal voltage:
   $E\angle\delta = Vt + jxd\ I$
2. Line current from the terminal to the infinite bus:
   $I = (Vt - V) / jx$
3. Substitute the numbers:
   $E\angle\delta = 1.1\angle\theta + j0.4\ (1.1\angle\theta - 1.0) / j1.0$
4. Simplify:
   $E\angle\delta = (1.54 \cos\theta - 0.4) + j1.54 \sin\theta$

SO FAR

$E\angle\delta = (1.54 \cos\theta - 0.4)$

$+ j1.54 \sin\theta$

STEADY-STATE STABILITY LIMIT · WORKED EXAMPLE

Solution (2): maximum power

1. At maximum power δ = 90°, so the real part of E is zero:
   1.54 cos θ = 0.4 → θ = 74.95°
2. |E| is then the imaginary part:
   |E| = 1.54 sin 74.95° = 1.487 pu
3. Maximum steady-state power:
   Pmax = |E||V| / (xd + x)
4. **= 1.487 × 1.0 / 1.4 = 1.062 pu**

ANSWER

Pmax ≈ 1.06 pu

Reached at θ = 74.95°, δ = 90°

EQUAL AREA CRITERION

- Used to study transient stability.
- Checks stability after a severe disturbance.
- Based on energy stored in the rotating mass.
- Disturbances include faults, load changes and loss of generation.
- Applicable to a single-machine infinite-bus or two-machine system.

Eg: When there is a sudden increase in mechanical input

- A1 is the accelerating area.
- A1 is usually enclosed by the initial operating point, the new equilibrium point, and the mechanical input
- A1 occurs because the input power increased suddenly before the rotor angle could change
- The excess mechanical power accelerated the rotor to the new equilibrium point (stable equilibrium angle)
- But due to momentum and excess energy available in the rotor, the rotor moves further ahead
- But as soon as the rotor crosses the new equilibrium angle, the electrical output exceeds mechanical input – so the rotor decelerates and dissipates the excess energy
- If the rotor moves beyond the unstable equilibrium angle, the rotor becomes unstable
- The area enclosed by the stable equilibrium point, unstable equilibrium point, and the mechanical input is the Area A2
- It represents the space available for the rotor to decelerate without reaching the unstable equilibrium point
- If the rotor could dissipate the excess energy before reaching the unstable equilibrium angle, then the system is stable
- If the initial excess energy is so much that the rotor has to go beyond the unstable equilibrium point to dissipate it, the machine becomes unstable

OBJECTIVE OF EQUAL AREA CRITERION

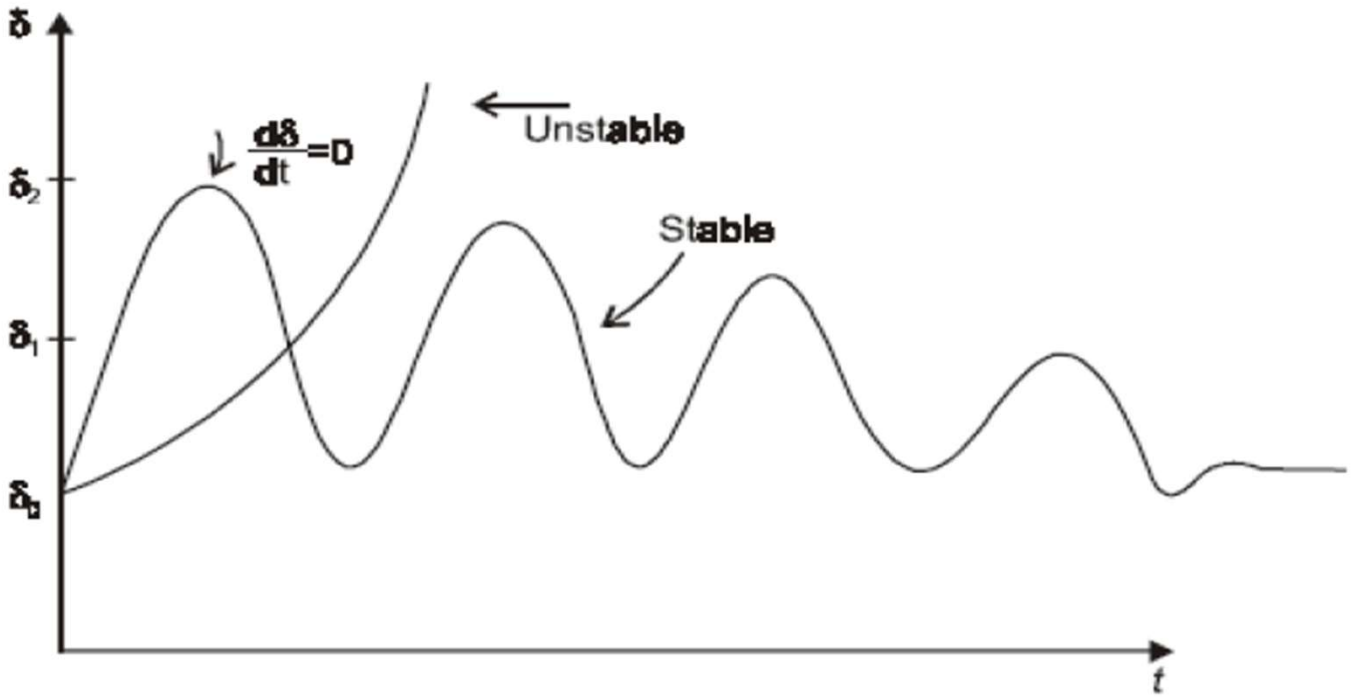
- Check whether synchronism is maintained.
- Compare accelerating and decelerating energy.
- Determine the maximum rotor angle.
- Predict stability without solving the swing equation directly.

Stability condition:

$$A_1 = A_2$$

A_1 – accelerating area

A_2 – decelerating area


δ
δ2
δ1
δ0
dδ/dt=0
Unstable
Stable
t

BASIS OF EQUAL AREA CRITERION

From the swing equation,

$$M\frac{d^2\delta}{dt^2} = P_a$$

and

$$P_a = P_i - P_e$$

Therefore,

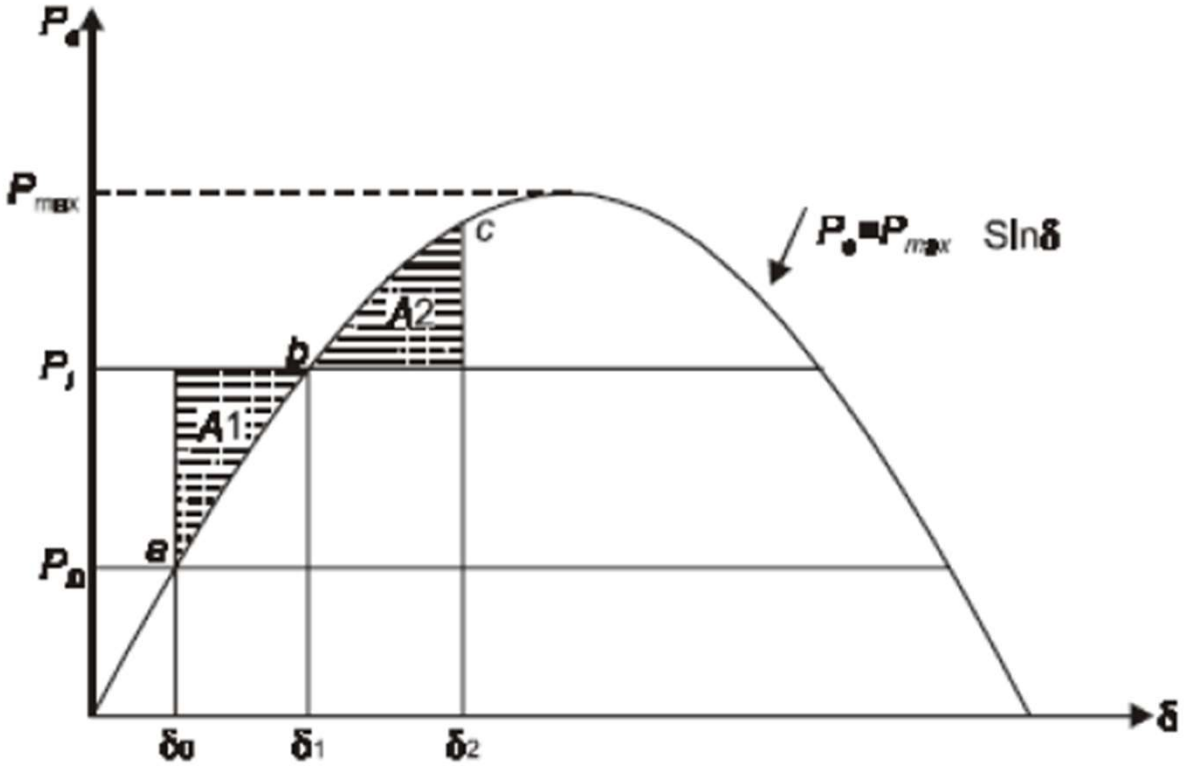
$$M\frac{d^2\delta}{dt^2} = P_i - P_e$$

P_i – mechanical input power

P_e – electrical output power

P_a – accelerating power

δ – rotor power angle



Initially, the generator operates at point a:

$$P_0 = P_e = P_{\max} \sin \delta_0$$

The mechanical input is suddenly increased from P_0 to P_i, while the electrical power-angle characteristic remains

$$P_e = P_{\max} \sin \delta$$

ACCELERATING AREA

When

$$P_i > P_e$$

then

$$P_a > 0$$

- Rotor accelerates.
- Rotor gains kinetic energy.
- Power angle increases.

Accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_1} (P_i - P_e)\, d\delta$$

A_1 represents energy gained by the rotor.

DECELERATING AREA

When

$$P_e > P_i$$

then

$$P_a < 0$$

- Rotor decelerates.
- Rotor loses kinetic energy.
- Rotor reaches a maximum angle.

Decelerating area:

$$A_2 = \int_{\delta_1}^{\delta_2} (P_e - P_i)\, d\delta$$

A_2 represents energy removed from the rotor.

For the system to be stable, the available decelerating area A_2 must be at least equal to the accelerating area A_1:

$$\boxed{A_2 \geq A_1}$$

More specifically:

- $A_2 > A_1$: sufficient decelerating capability → stable.
- $A_2 = A_1$: critical/limiting stable condition.
- $A_2 < A_1$: insufficient deceleration → unstable, rotor loses synchronism.

So the key condition is:

$$\boxed{\text{Stable: } A_2 \geq A_1}$$

Here A_1 represents kinetic energy gained during acceleration, while A_2 represents the available capability to remove that energy.

STABILITY CONDITION

For stable operation,

$$A_1 = A_2$$

Hence,

$$\int_{\delta_0}^{\delta_1} (P_i - P_e) d\delta = \int_{\delta_1}^{\delta_2} (P_e - P_i) d\delta$$

- Accelerating energy = decelerating energy.
- Rotor reaches a maximum angle.
- Rotor then swings back.
- Synchronism is maintained.

Stable and Unstable Equilibrium Angles

For a constant mechanical input P_i, the line P_i intersects the power-angle curve at two points:

$$P_i = P_{\max} \sin \delta$$

- δ_1 – stable equilibrium angle
  - First intersection with the power-angle curve.
  - Small disturbances tend to return the rotor toward this operating point.
- δ_u – unstable equilibrium (critical) angle
  - Second intersection with the power-angle curve.
  - It is the critical angle because beyond δ_u, $P_i > P_e$, so the rotor accelerates further instead of returning toward the stable operating point.
  - Hence, crossing δ_u results in loss of synchronism.

Since

$$\sin \delta_1 = \sin \delta_u$$

$$\boxed{\delta_u = 180^\circ - \delta_1}$$

Let δ_m be the maximum rotor angle reached during the swing:

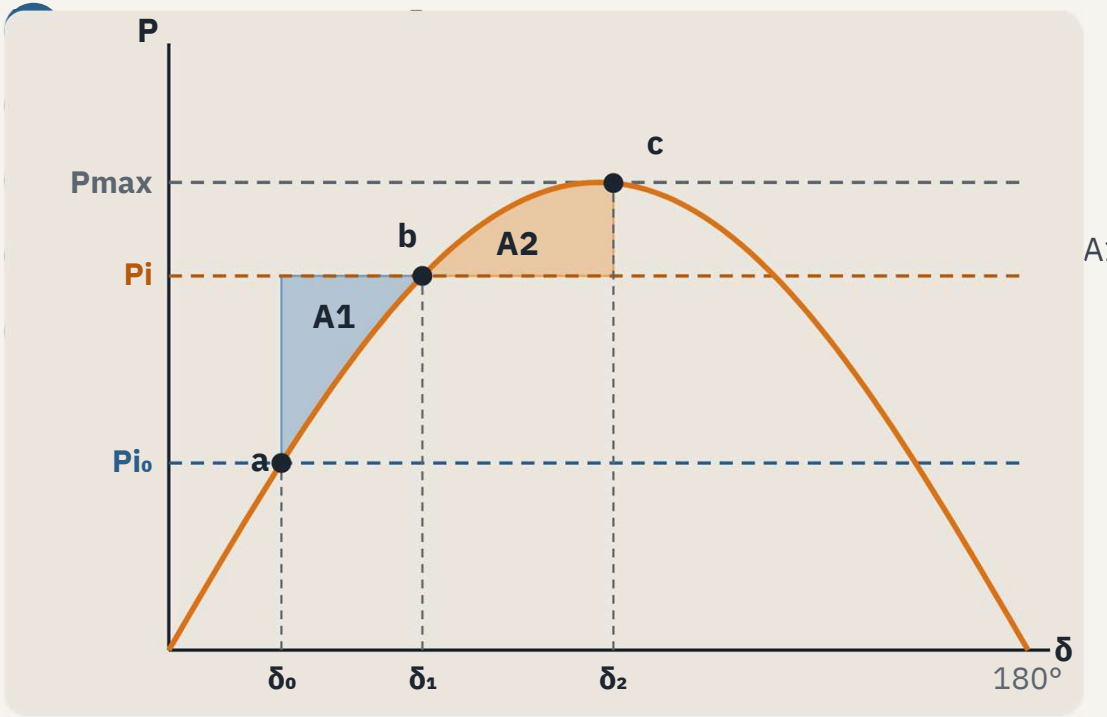
$$\boxed{\delta_m < \delta_u \Rightarrow \text{Stable}}$$

$$\boxed{\delta_m = \delta_u \Rightarrow \text{Critical stability limit}}$$

$$\boxed{\delta_m > \delta_u \Rightarrow \text{Loss of synchronism}}$$

EQUAL AREA CRITERION · APPLICATION 1

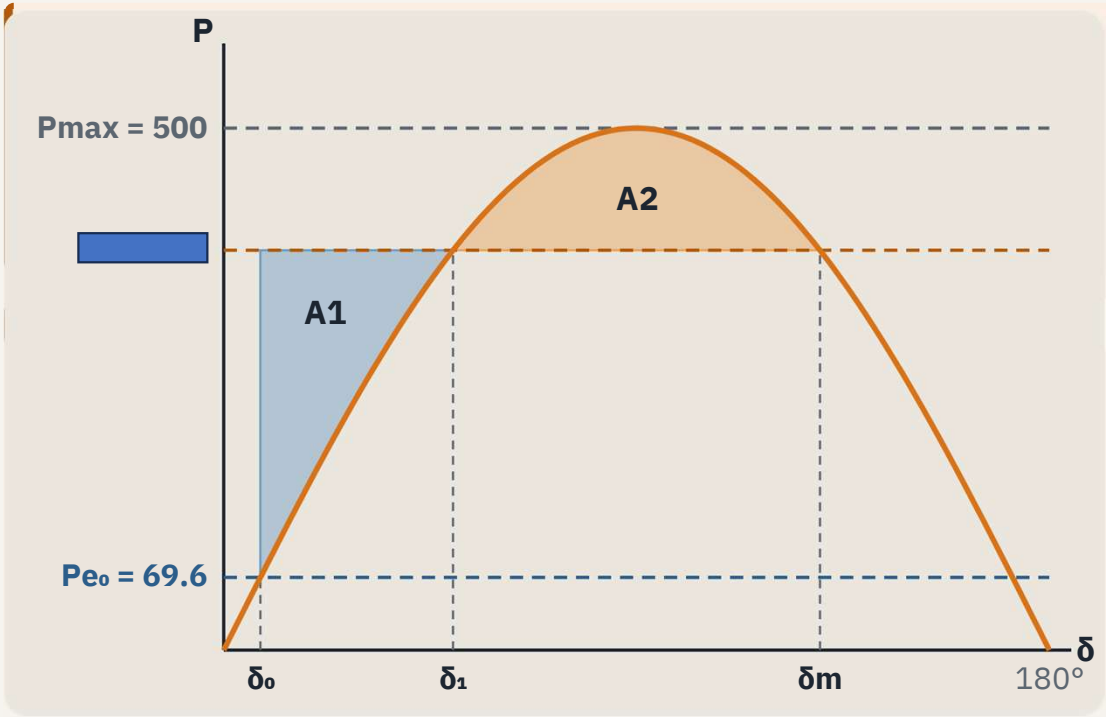
Sudden increase in input power



A1.

EQUAL AREA CRITERION · WORKED EXAMPLE

Example: how much extra input is safe?



$$P_e = 500 \sin \delta \text{ MW}$$

Initial operating angle:

$$\delta_0 = 8^\circ$$

Initial power:

$$P_{m0} = P_{e0} = 500 \sin 8^\circ = 69.6 \text{ MW}$$

Mechanical input is suddenly increased from 69.6 MW to an unknown value P_i.

At the new intersection with the power-angle curve:

$$P_i = 500 \sin \delta_1$$

Unknowns:

$$P_i, \quad \delta_1$$

At the limiting stable condition:

$$\delta_m = \pi - \delta_1$$

Goal: Find the maximum P_i for which the system remains stable.

Rotor accelerates, giving accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_1} (P_i - 500 \sin \delta)\, d\delta$$

From δ_1 to δ_m:

$$P_e > P_i$$

Rotor decelerates, giving decelerating area:

$$A_2 = \int_{\delta_1}^{\delta_m} (500 \sin \delta - P_i)\, d\delta$$

At the limiting stable condition:

$$\boxed{A_1 = A_2}$$

Also,

$$P_i = 500 \sin \delta_1$$

and

$$\delta_m = \pi - \delta_1$$

Applying

$$A_1 = A_2$$

gives

$$P_i(\delta_m - \delta_0) = 500(\cos\delta_0 - \cos\delta_m)$$

Substitute

$$P_i = 500\sin\delta_1$$

and

$$\delta_m = \pi - \delta_1$$

Since

$$\cos(\pi - \delta_1) = -\cos\delta_1$$

we get

$$(\pi - \delta_1 - \delta_0)\sin\delta_1 - \cos\delta_1 - \cos\delta_0 = 0$$

For

$$\delta_0 = 8^\circ = 0.1396 \text{ rad}$$

$$\boxed{(3.002 - \delta_1)\sin\delta_1 - \cos\delta_1 - 0.9903 = 0}$$

SOLUTION BY TRIAL AND ERROR

The equal-area equation is

$$(3.002 - \delta_1)\sin\delta_1 - \cos\delta_1 - 0.9903 = 0$$

This is a transcendental equation and is solved by TRIAL AND ERROR.

The new stable operating angle must lie between

$$8^\circ < \delta_1 < 90^\circ$$

Start the trial-and-error process with the midpoint:

$$\delta_1^{(1)} = \frac{8^\circ + 90^\circ}{2} = \boxed{49^\circ}$$

Convert each trial angle to radians before substitution.

Trial 1:

$$\delta_1 = 49^\circ = 0.8552\ \text{rad}$$

Substitution shows that the required root is slightly greater than 49°.

Further trial and error gives

$$\boxed{\delta_1 \approx 50.1^\circ}$$

Solving

$$(3.002 - \delta_1)\sin\delta_1 - \cos\delta_1 - 0.9903 = 0$$

gives approximately

$$\boxed{\delta_1 \approx 50.1^\circ}$$

Therefore,

$$\delta_m = 180^\circ - 50.1^\circ$$

$$\boxed{\delta_m \approx 129.9^\circ}$$

Now calculate the previously unknown mechanical input:

$$P_i = 500\sin\delta_1$$

$$P_i = 500\sin 50.1^\circ$$

$$\boxed{P_i \approx 383.5\ \text{MW}}$$

Maximum permissible sudden increase:

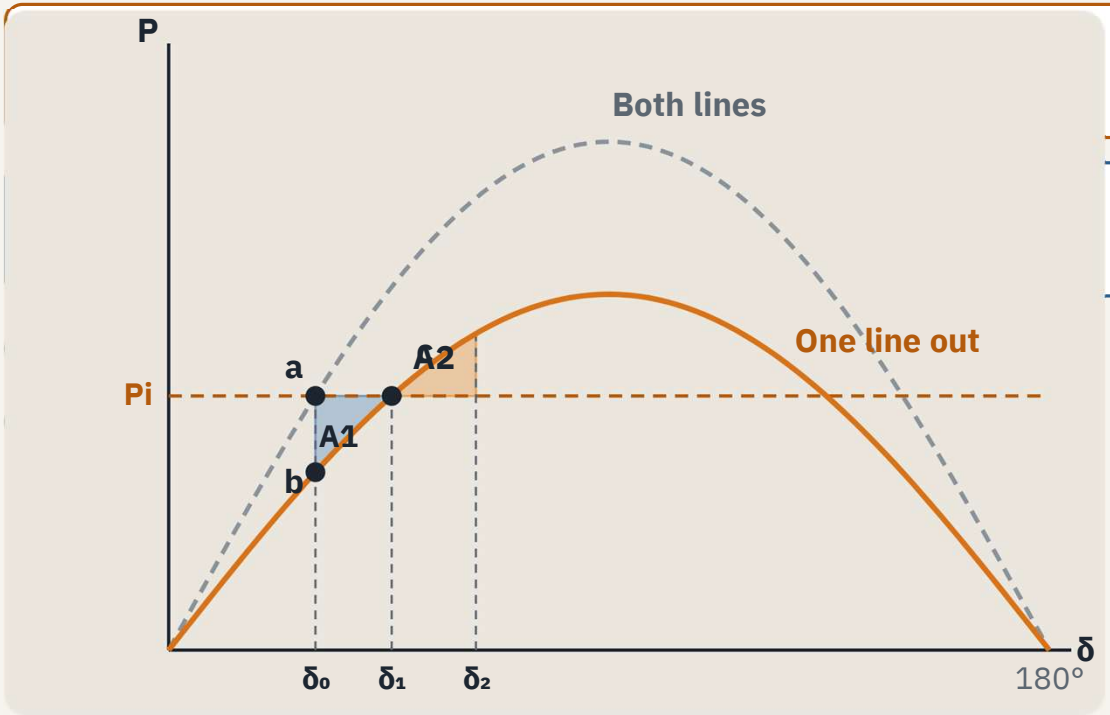
$$\Delta P = P_i - P_{m0}$$

$$= 383.5 - 69.6$$

$$\boxed{\Delta P \approx 313.9\ \text{MW}}$$

EQUAL AREA CRITERION · APPLICATION 2

Sudden loss of one parallel line



Sudden Loss of One Parallel Line – Initial Condition

- Initially, both parallel transmission lines are in service.
- Parallel lines provide a lower equivalent transfer reactance X.
- Hence, the electrical power-transfer capability is high:

$$P_{e1} = P_{\max 1} \sin \delta$$

- The generator operates at point a, with rotor angle δ_0.
- At steady state:

$$P_i = P_{e1}$$

- Therefore, accelerating power is zero:

$$P_a = P_i - P_e = 0$$

Sudden Loss of One Parallel Line

- One of the parallel lines is suddenly disconnected.
- Equivalent transfer reactance increases:

$$X_{\text{after}} > X_{\text{before}}$$

- Since

$$P_{\max} = \frac{EV}{X}$$

the maximum transferable power decreases.

- The power-angle characteristic changes from the upper dashed curve to the lower curve.
- Rotor angle cannot change instantaneously, so it remains at δ_0.
- Hence, the operating point shifts vertically from a to b.

At point b:

$$P_i > P_e$$

Therefore,

$$P_a = P_i - P_e > 0$$

and the rotor starts accelerating.

Accelerating Area A_1

- As the rotor accelerates, the rotor angle increases from δ_0 to δ_1.
- The energy gained by the rotor is represented by the accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_1} (P_i - P_e)\, d\delta$$

- At δ_1:

$$P_i = P_e$$

- Therefore, accelerating power becomes zero.
- However, the rotor has gained kinetic energy and continues to move beyond δ_1.

Decelerating Area A_2

- Beyond δ_1:

$$P_e > P_i$$

- Therefore,

$$P_a = P_i - P_e < 0$$

and the rotor starts decelerating.

- The energy removed from the rotor is represented by:

$$A_2 = \int_{\delta_1}^{\delta_2} (P_e - P_i)\, d\delta$$

- At the maximum rotor angle δ_2, the rotor has lost the kinetic energy gained during acceleration.

For stable operation:

$$\boxed{A_1 = A_2}$$

A synchronous generator is connected to an infinite bus through two parallel transmission lines. With both lines in service, the maximum power transfer is

$$P_{\max 1} = 2.0\ pu$$

The generator is initially operating with a constant mechanical input of

$$P_m = 0.8\ pu$$

One of the parallel transmission lines is suddenly disconnected, reducing the maximum power transfer to

$$P_{\max 2} = 1.2\ pu$$

Using the equal area criterion:

(a) Determine the initial rotor angle δ_0.

(b) Determine the new steady-state equilibrium angle δ_1.

(c) Determine the maximum rotor swing angle following the line outage and assess the transient stability of the system.

(a) Initial rotor angle, δ_0

Before the line outage, both lines are in service and the generator is in steady state:

$$P_m = P_{\max 1} \sin \delta_0$$

$$0.8 = 2.0 \sin \delta_0$$

$$\sin \delta_0 = 0.4$$

Therefore,

$$\boxed{\delta_0 = 23.58^\circ}$$

(b) New steady-state equilibrium angle, δ_1

After one line is disconnected:

$$P_{\max 2} = 1.2\ pu$$

At the new equilibrium point:

$$P_m = P_{\max 2} \sin \delta_1$$

$$0.8 = 1.2 \sin \delta_1$$

$$\sin \delta_1 = 0.6667$$

Therefore,

$$\boxed{\delta_1 = 41.81^\circ}$$

The corresponding unstable equilibrium angle is:

$$\delta_u = 180^\circ - \delta_1$$

$$\boxed{\delta_u = 138.19^\circ}$$

Using the equal area criterion:

$$A_1 = A_2$$

Accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_1} (0.8 - 1.2\sin\delta)d\delta$$

Decelerating area:

$$A_2 = \int_{\delta_1}^{\delta_2} (1.2\sin\delta - 0.8)d\delta$$

Equating and simplifying:

$$0.8(\delta_2 - \delta_0) = 1.2(\cos\delta_0 - \cos\delta_2)$$

Convert the initial angle to radians:

$$\delta_0 = 23.58^\circ = 0.4115\ rad$$

Therefore,

$$0.8(\delta_2 - 0.4115) = 1.2(0.9165 - \cos\delta_2)$$

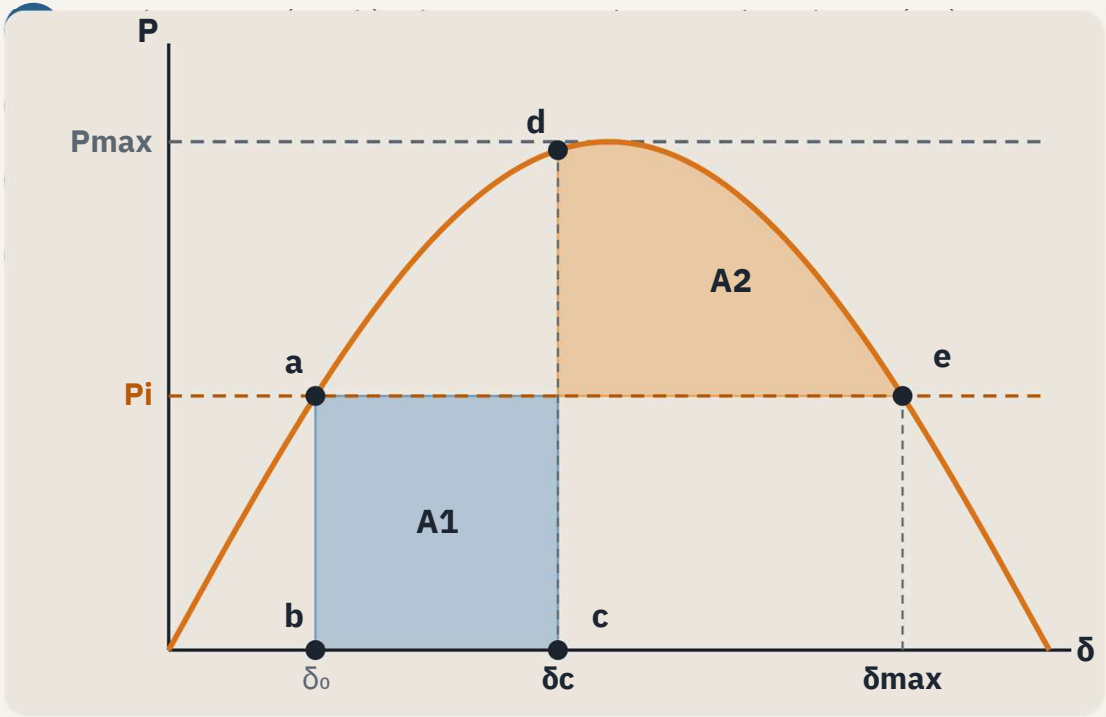
Solving gives:

$$\boxed{\delta_2 \approx 61.5^\circ}$$

$\delta_2 < \delta_u$: So the System is stable

EQUAL AREA CRITERION · APPLICATION 3

Three-phase fault and fault clearing



Three-Phase Fault and Fault Clearing – Equal Area Criterion

- Initially, the generator operates at point a with rotor angle δ_0.

$$P_i = P_e$$

- When a severe three-phase fault occurs, the electrical power output drops approximately to zero:

$$P_e \approx 0$$

- Mechanical input P_i remains unchanged. Therefore,

$$P_a = P_i - P_e \approx P_i > 0$$

- The rotor accelerates and the rotor angle increases from δ_0 toward the clearing angle δ_c.

During the Fault – Accelerating Area

- The fault persists from δ_0 to δ_c.
- Since $P_e \approx 0$, the generator continuously accelerates.
- The kinetic energy gained by the rotor is represented by area A_1:

$$A_1 = \int_{\delta_0}^{\delta_c} (P_i - P_e)\, d\delta$$

For $P_e \approx 0$:

$$\boxed{A_1 = P_i(\delta_c - \delta_0)}$$

- δ_c is called the fault clearing angle — the rotor angle at the instant the fault is cleared.

After Fault Clearing – Decelerating Area

- At δ_c, the protection system clears the fault.
- In the figure, the original power-angle characteristic is restored:

$$P_e = P_{\max} \sin \delta$$

- Immediately after clearing,

$$P_e > P_i$$

so the rotor begins to decelerate.

- However, the rotor has already gained kinetic energy and therefore continues to increase its angle beyond δ_c.
- The decelerating energy is represented by area A_2:

$$A_2 = \int_{\delta_c}^{\delta_{\max}} (P_e - P_i)\, d\delta$$

- At $\delta_{\max}$, the excess kinetic energy has been removed and the rotor reaches its maximum swing angle.

For the generator to remain stable, sufficient decelerating area must be available to remove the kinetic energy gained during the fault.

$$\boxed{A_2 \geq A_1}$$

At the critical stability limit:

$$\boxed{A_1 = A_2}$$

and the maximum rotor swing reaches the unstable equilibrium angle:

$$\boxed{\delta_{\max} = \delta_u}$$

where

$$\boxed{\delta_u = \pi - \delta_0}$$

for the restored pre-fault power-angle curve shown in the figure.

The corresponding value of δ_c is called the critical clearing angle:

$$\boxed{\delta_{cc} = \text{maximum fault-clearing angle for maintaining stability}}$$

Therefore:

$$\boxed{\delta_c < \delta_{cc} \Rightarrow \text{Stable}}$$

$$\boxed{\delta_c > \delta_{cc} \Rightarrow \text{Unstable}}$$

Critical Clearing Time

Critical clearing time t_{cc} is the time taken by the rotor angle to increase from the initial angle δ_0 to the critical clearing angle δ_{cc}.

For a severe 3-phase fault:

$$P_e \approx 0$$

Hence, accelerating power is constant:

$$P_a = P_m - P_e = P_m$$

From the swing equation,

$$\frac{2H}{\omega_s}\frac{d^2\delta}{dt^2} = P_m$$

Therefore,

$$\frac{d^2\delta}{dt^2} = \frac{\omega_s P_m}{2H}$$

Since the rotor has zero initial speed deviation, integrating twice:

$$\delta - \delta_0 = \frac{\omega_s P_m}{4H} t^2$$

At critical clearing,

$$\delta = \delta_{cc}, \qquad t = t_{cc}$$

Therefore,

$$\boxed{t_{cc} = \sqrt{\frac{4H(\delta_{cc} - \delta_0)}{\omega_s P_m}}}$$

A synchronous generator connected to an infinite bus has a maximum power transfer of

$$P_{\max} = 2.0\ pu$$

and operates with a mechanical input

$$P_m = 0.8\ pu$$

The inertia constant of the generator is $H = 5$ s and the system frequency is 50 Hz. A three-phase fault occurs, reducing the electrical output power during the fault to zero. After clearing, the original power-angle characteristic is restored.

Using the Equal Area Criterion:

(a) Determine the initial rotor angle δ_0.

(b) Determine the critical clearing angle δ_{cc}.

(c) Determine the critical clearing time t_{cc}.

(a) Initial Rotor Angle

Before the fault, the generator operates in steady state:

$$P_i = P_{\max} \sin \delta_0$$

$$0.8 = 2.0 \sin \delta_0$$

Therefore,

$$\sin \delta_0 = 0.4$$

$$\boxed{\delta_0 = 23.58^\circ}$$

(b) Unstable Equilibrium Angle

Since the original power-angle characteristic is restored after fault clearing,

$$P_i = P_{\max} \sin \delta$$

The unstable equilibrium angle is:

$$\delta_u = 180^\circ - \delta_0$$

Therefore,

$$\boxed{\delta_u = 156.42^\circ}$$

At the critical stability limit,

$$\boxed{\delta_{\max} = \delta_u}$$

(c) Critical Clearing Angle

During the three-phase fault,

$$P_e = 0$$

Therefore, the accelerating area is:

$$A_1 = P_i(\delta_{cc} - \delta_0)$$

After the fault is cleared, the decelerating area is:

$$A_2 = \int_{\delta_{cc}}^{\delta_u} (P_{\max} \sin\delta - P_i) d\delta$$

At the critical stability limit:

$$A_1 = A_2$$

Hence,

$$P_i(\delta_{cc} - \delta_0) = P_{\max}(\cos\delta_{cc} - \cos\delta_u) - P_i(\delta_u - \delta_{cc})$$

Simplifying:

$$P_i(\delta_u - \delta_0) = P_{\max}(\cos\delta_{cc} - \cos\delta_u)$$

Therefore,

$$\boxed{\cos\delta_{cc} = \cos\delta_u + \frac{P_i}{P_{\max}}(\delta_u - \delta_0)}$$

Angles must be in radians:

$$\delta_0 = 0.4115\ rad$$

$$\delta_u = 2.7301\ rad$$

Substituting:

$$\cos\delta_{cc} = \cos(2.7301) + \frac{0.8}{2.0}(2.7301 - 0.4115)$$

$$\cos\delta_{cc} \approx 0.011$$

Therefore,

$$\boxed{\delta_{cc} \approx 89.4^\circ}$$

$$\boxed{t_{cc} = \sqrt{\frac{4H(\delta_{cc} - \delta_0)}{\omega_s P_m}}}$$

Given:

$$H = 5 \text{ s}, \qquad f = 50 \text{ Hz}, \qquad \omega_s = 2\pi f = 314.16 \text{ rad/s}$$

$$\delta_0 = 23.58^\circ = 0.4115 \text{ rad}$$

$$\delta_{cc} = 89.4^\circ = 1.5603 \text{ rad}$$

Therefore,

$$t_{cc} = \sqrt{\frac{4(5)(1.5603 - 0.4115)}{314.16(0.8)}}$$

$$\boxed{t_{cc} \approx 0.302 \text{ s}}$$

Thus, the fault must be cleared within approximately 0.30 s to maintain transient stability.

EQUAL AREA CRITERION · APPLICATION 3

Critical clearing angle

The **critical clearing angle** δcr is the largest clearing angle for which the system stays stable.

using $\delta_u = \pi - \delta_0$ and $P_m / P_{\max} = \sin \delta_0$

$$\cos \delta_{cr} = (\pi - 2\delta_0) \sin \delta_0 - \cos \delta_0$$

How it is found

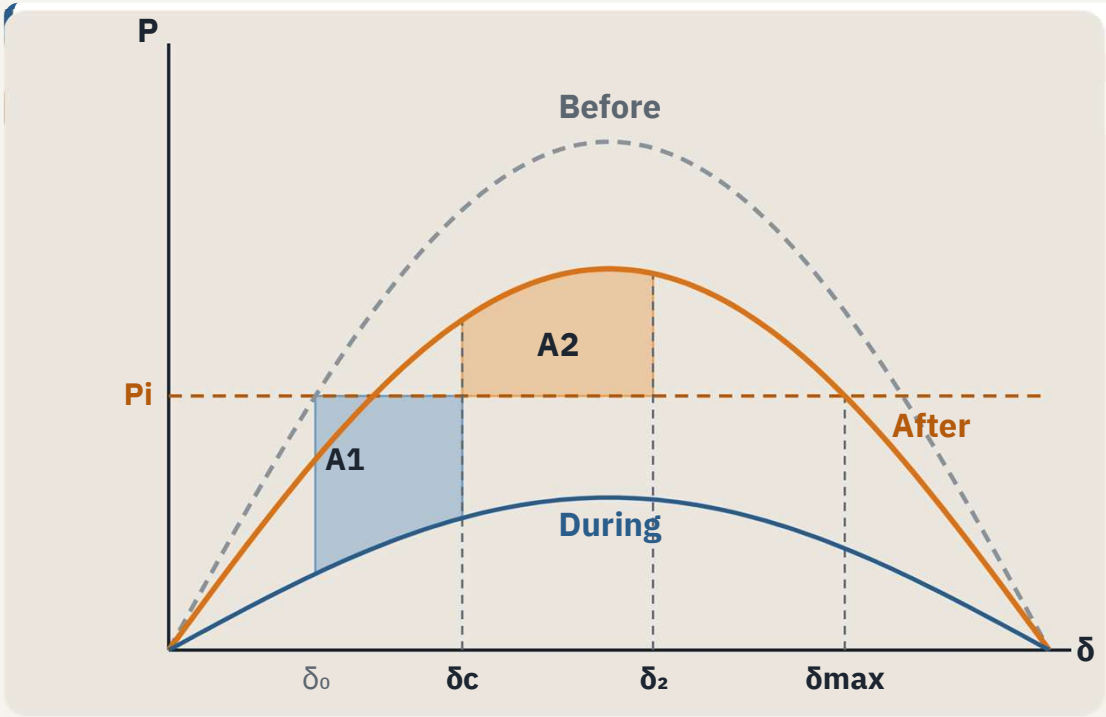
Set A1 = A2 with δmax = π − δ₀ and Pi = Pmax sin δ₀.

Units

All angles in the formula are in radians.

EQUAL AREA CRITERION · APPLICATION 4

Fault in the middle of a line



EQUAL AREA CRITERION · APPLICATION 4

Critical clearing angle for a mid-line fault

$$\cos \delta_{cr} = \frac{P_i (\delta_{max} - \delta_0) - P_{maxII} \cos \delta_0 + P_{maxIII} \cos \delta_{max}}{P_{maxIII} - P_{maxII}}$$

Largest swing allowed

$\delta_{max} = \pi - \sin^{-1}(P_i / P_{maxIII})$

PmaxII and PmaxIII

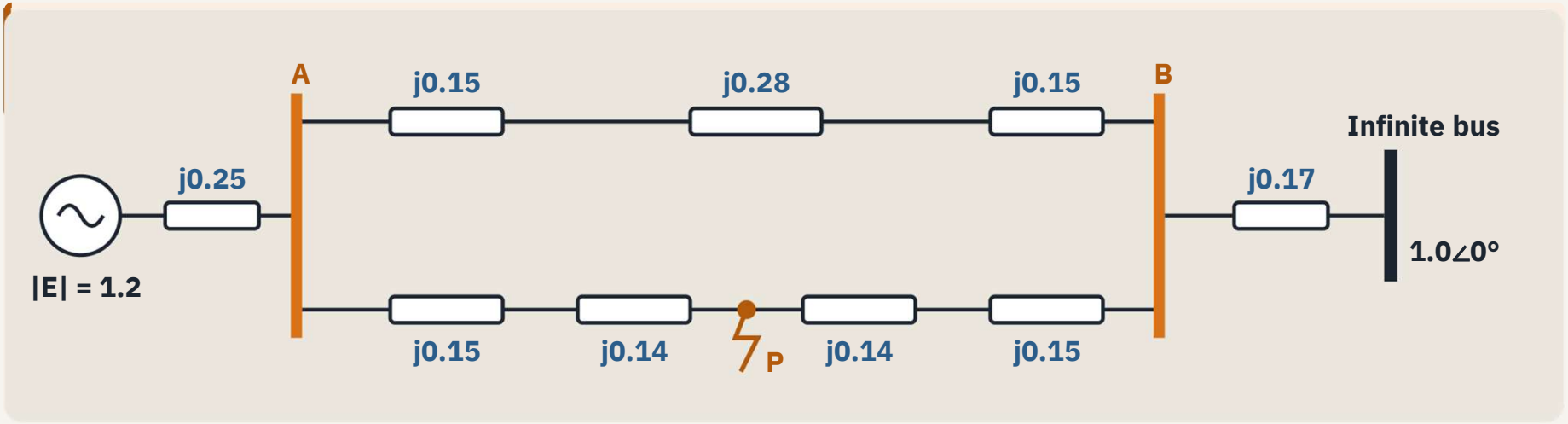
Peaks of the during-fault and after-clearing curves.

Units

Angles in radians. For degrees, multiply the Pi term by π/180.

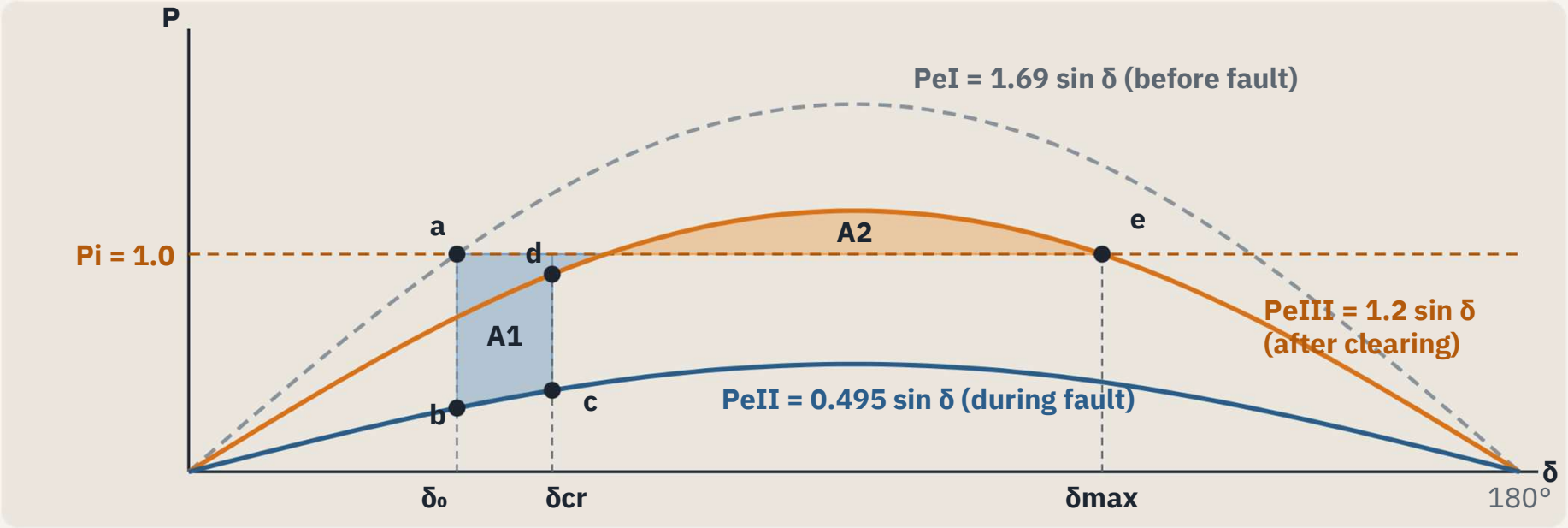
EQUAL AREA CRITERION · WORKED EXAMPLE

Example: critical clearing angle



EQUAL AREA CRITERION · WORKED EXAMPLE

Equal area curves for this example



WORKED EXAMPLE · TRANSFER REACTANCE

Before the fault: transfer reactance XI

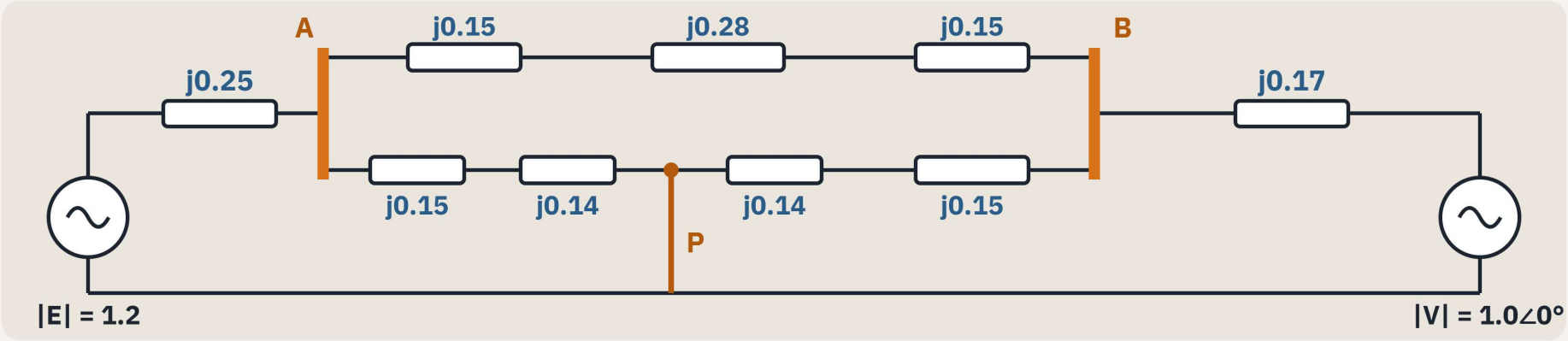


$$XI = 0.25 + 0.29 + 0.17 = 0.71 \text{ pu}$$

$$PeI = \frac{|E|\,|V|}{XI} \sin\delta = \frac{1.2 \times 1.0}{0.71} \sin\delta = 1.69 \sin\delta$$

WORKED EXAMPLE · TRANSFER REACTANCE

During the fault: P is shorted to neutral



Fault at P

A three-phase fault joins P to the neutral (zero voltage).

Lower line splits

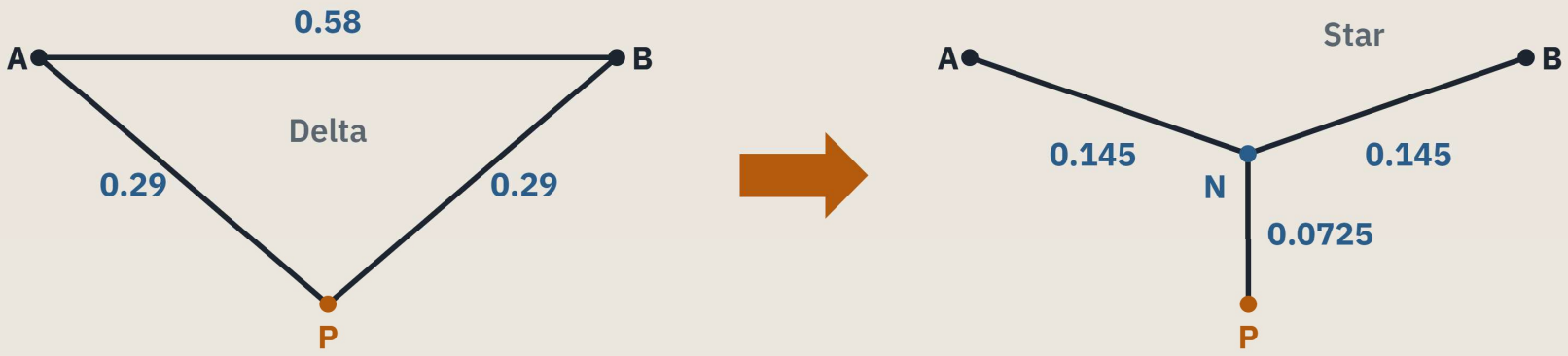
A to P: 0.15 + 0.14 = 0.29 pu. P to B: 0.14 + 0.15 = 0.29 pu.

A delta forms

A–B = 0.58, A–P = 0.29, B–P = 0.29 pu.

WORKED EXAMPLE · TRANSFER REACTANCE

During the fault: delta to star



Each star arm = (product of the two delta sides meeting at that node) ÷ (sum of all three sides = 1.16)

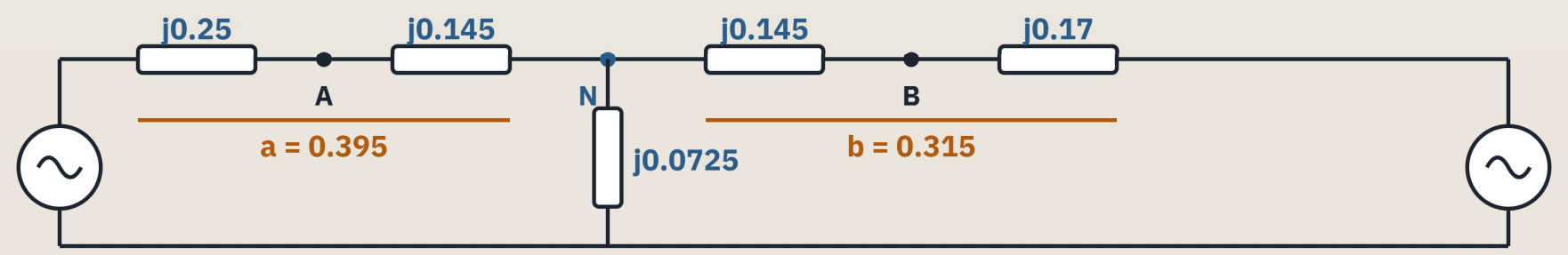
XA = 0.58 × 0.29 / 1.16 = 0.145

XB = 0.58 × 0.29 / 1.16 = 0.145

XP = 0.29 × 0.29 / 1.16 = 0.0725

WORKED EXAMPLE · TRANSFER REACTANCE

During the fault: the T network



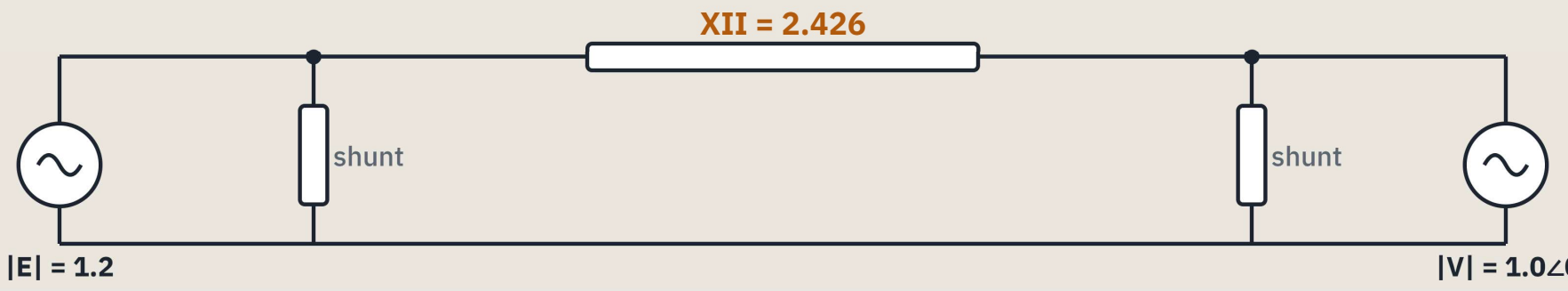
E side
a = 0.25 + 0.145 = 0.395 pu

V side
b = 0.145 + 0.17 = 0.315 pu

Shunt to neutral
c = 0.0725 pu

WORKED EXAMPLE · TRANSFER REACTANCE

During the fault: star to delta gives XII



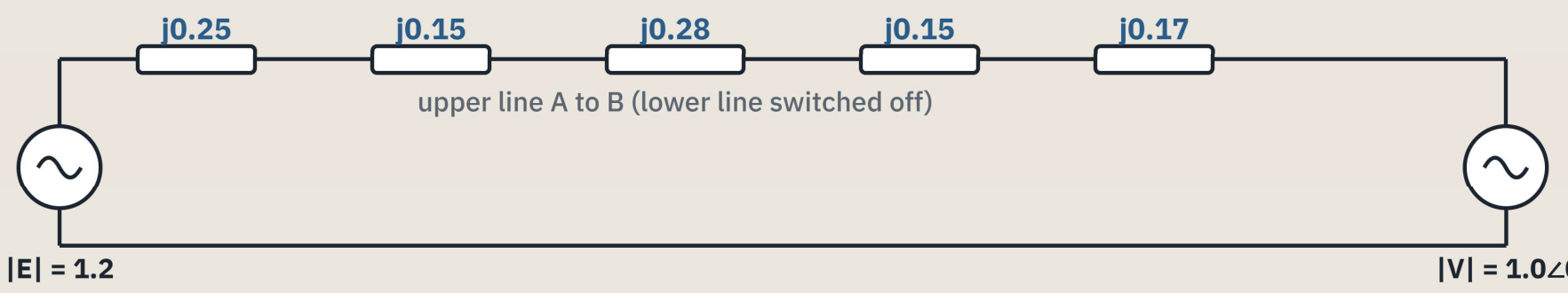
$$XII = \frac{ab + bc + ca}{c} = \frac{0.395\times0.315 + 0.315\times0.0725 + 0.0725\times0.395}{0.0725} = 2.426$$

$$PeII = \frac{1.2 \times 1.0}{2.426} \sin\delta = 0.495 \sin\delta$$

The shunt branches go to neutral, so they carry no power from E to V.

WORKED EXAMPLE · TRANSFER REACTANCE

After the fault: transfer reactance XIII



$$X_{III} = 0.25 + 0.15 + 0.28 + 0.15 + 0.17 = 1.0 \text{ pu}$$

$$P_{eIII} = \frac{1.2 \times 1.0}{1.0} \sin\delta = 1.2 \sin\delta$$

$X_{III} > X_I$, so the post-fault curve is lower than the pre-fault curve.

EQUAL AREA CRITERION · WORKED EXAMPLE

Step 1: the three power angle curves

Condition	Transfer reactance (pu)	Power curve
Before fault	XI = 0.25 + 0.17 + (0.58 ‖ 0.58) = 0.71	PeI = 1.69 sin δ
During fault	XII = 2.426	PeII = 0.495 sin δ
After clearing	XIII = 0.25 + 0.15 + 0.28 + 0.15 + 0.17 = 1.0	PeIII = 1.2 sin δ

Finding XII (during the fault)

Delta–star turns the faulted section into arms 0.145, 0.145 and 0.0725 pu. Star–delta then gives XII = (0.395 × 0.315 + 0.315 × 0.0725 + 0.0725 × 0.395) / 0.0725 = 2.426 pu.

EQUAL AREA CRITERION · WORKED EXAMPLE

Step 2: initial and maximum angles

1. Initial angle before the fault
   $1.0 = 1.69 \sin \delta_0 \rightarrow \delta_0 = 36.3° = 0.633$ rad
2. Largest swing allowed after clearing
   $\delta_{max} = \pi - \sin^{-1}(1.0 / 1.2)$
3. **$= 2.156$ rad (123.6°)**

SO FAR

$\delta_0 = 0.633$ rad

$\delta_{max} = 2.156$ rad

EQUAL AREA CRITERION · WORKED EXAMPLE

write the two areas as integrals

1. A1: the rotor accelerates on PeII until δcr, then on PeIII until δf = 56.4°, where PeIII = Pi:
   $A1 = \int(P_i - P_{maxII} \sin\delta)\, d\delta\ [\delta_0 \to \delta_{cr}] + \int(P_i - P_{maxIII} \sin\delta)\, d\delta\ [\delta_{cr} \to \delta_f]$

2. A2: the rotor decelerates from δf to δmax:
   $A2 = \int(P_{maxIII} \sin\delta - P_i)\, d\delta\ [\delta_f \to \delta_{max}]$

3. Integrate and set A1 = A2. All the δf terms cancel, leaving:
   $P_i(\delta_{max} - \delta_0) + P_{maxII}(\cos\delta_{cr} - \cos\delta_0) + P_{maxIII}(\cos\delta_{max} - \cos\delta_{cr}) = 0$

4. Solve for cos δcr (the formula from Step 3):
   $\cos\delta_{cr} = [P_i(\delta_{max} - \delta_0) - P_{maxII}\cos\delta_0 + P_{maxIII}\cos\delta_{max}] / (P_{maxIII} - P_{maxII}) = 0.654$

EQUAL AREA CRITERION · WORKED EXAMPLE

Step 3: critical clearing angle

1. Substitute in the mid-line fault formula

   $\cos \delta_{cr} = [1.0 \times (2.156 - 0.633) - 0.495 \cos 36.3° + 1.2 \cos 123.6°] / (1.2 - 0.495)$

2. **$= (1.523 - 0.399 - 0.663) / 0.705 = 0.654$**

3. Take the inverse cosine

   $\delta_{cr} = \cos^{-1}(0.654) = 49.2°$

ANSWER

$\delta_{cr} \approx 49.2°$

EQUAL AREA CRITERION · WORKED EXAMPLE

Step 5: calculate the accelerating area A1

Angles in radians: δ_0 = 0.6331 (36.3°), δ_{cr} = 0.8580 (49.2°), δ_f = 0.9851 (56.4°, where PeIII = Pi), δ_{max} = 2.1565 (123.6°).

1. Part 1, fault on ($\delta_0 \rightarrow \delta_{cr}$, curve PeII):
   A1a = 1.0 × (0.8580 − 0.6331) + 0.495 × (0.6539 − 0.8062)
2. **= 0.2249 − 0.0754 = 0.1495**
3. Part 2, after clearing ($\delta_{cr} \rightarrow \delta_f$, curve PeIII still below Pi):
   A1b = 1.0 × (0.9851 − 0.8580) + 1.2 × (0.5528 − 0.6539)
4. **= 0.1271 − 0.1214 = 0.0057**

ANSWER

A1 = 0.1495 + 0.0057

= 0.155 pu·rad

EQUAL AREA CRITERION · WORKED EXAMPLE

Step 6: calculate the decelerating area A2

From the previous slide: δf = 0.9851 rad (56.4°), δmax = 2.1565 rad (123.6°), A1 = 0.155 pu·rad.

1. Decelerating region (δf → δmax, curve PeIII above Pi):
   A2 = 1.2 × (cos δf − cos δmax) − 1.0 × (δmax − δf)
2. **= 1.2 × (0.5528 + 0.5528) − 1.0 × (2.1565 − 0.9851)**
3. **= 1.3267 − 1.1714 = 0.1553**
4. Compare with A1:
   A1 = A2 = 0.155 pu·rad

ANSWER

A2 = 0.155 pu·rad

A1 = A2 ✓

δcr = 49.2° confirmed

δ_0 = 36.3° · δcr = 49.2° · δmax = 123.6° · A1 = A2 = 0.155 pu·rad

IMPROVING STABILITY

Improving stability (1): raise the power curve

$$P = \frac{|E|\,|V|}{X}\sin\delta$$

Raising |E| or |V|, or lowering X, lifts the curve and raises Pmax.

Higher voltage
Power transfer rises with the square of the voltage level.

More parallel lines
Lines in parallel reduce the transfer reactance.

Series capacitors
Cancel part of the line reactance.

Low-reactance equipment
Bundled conductors, low-leakage transformers.

SVC and FACTS
Hold bus voltages up during disturbances.

Fast excitation
Raises |E| quickly when voltage dips.

IMPROVING STABILITY

Improving stability (2): smaller A1, larger A2

Goal: less energy gained during the fault, more removed after it.

Fast fault clearing

Fast relays and breakers: smaller δc, smaller A1.

Fast reclosing

Restores the line and the higher curve quickly.

Single-pole switching

Healthy phases keep carrying power.

High inertia, low X′d

δ rises more slowly; the curve is higher.

Fast valving

Cuts turbine power during the fault.

Braking resistor

Absorbs surplus power, slowing the rotor.

SUMMARY

Key formulas

Power angle curve

$P = (|E||V| / X) \sin \delta$

Steady-state limit

P_{max} at $\delta = 90°$; $S_p = P_{max} \cos \delta$

Inertia constant

H = energy / MVA; $M = GH / \pi f$

Swing equation

$(H / \pi f)\, d^2\delta/dt^2 = P_m - P_e$

Equal area criterion

Stable if $A1 = A2$ before δ_{max}

Critical clearing

$\cos \delta_{cr} = (\pi - 2\delta_0) \sin \delta_0 - \cos \delta_0$